

SF2321 Applied Statistical Analysis (second cycle)

COURSE GUIDE FALL 2025

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For general questions about the course, contact Maria Solevid. For questions about course work related to a specific assignment, contact the responsible instructor (see course work overview on page 4). For questions about registration, course credits etc, contact course administrator Sandra Grahn (master@pol.gu.se).

The updated schedule and additional information are available at the course's Canvas page. You will be informed about any changes, additions etc. through notifications on Canvas. In other words, please check the Canvas page, and your Canvas connected email inbox, regularly during the course.

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Overview

This course provides students with a comprehensive understanding of the design and application of methods for causal inference using the most common regression techniques featured in social science research and commercial analyses and reports. The course also teaches students how to conduct statistical analysis using Stata, and how to interpret the empirical results from statistical analyses. We further emphasize students' learning by letting students independently choose research questions, data sets and variables for several of the assignments.

The course is divided into three main parts. The **first** focuses on theory and research design for causal inference (assignment 1) but also offers recap lectures and an introduction to the statistical software Stata. The aim with the lecture and the subsequent Stata tutorial is to provide an opportunity to revise basic statistical analysis, to start get to know Stata and to get everyone up to speed. The **second**, and largest part of the course, is devoted to lectures, teacher-led tutorials, and assignments on linear regression and the empirical application of statistical research methods for causal inference (assignment 2-6).

The **third** part of the course is focused on an independent research paper, the Method thesis (assignment 7), in which learned skills are put into practice and the design and statistical methods for causal inference are applied on a self-chosen research problem and data material. You will be assigned one of the course instructors as your advisor. The actual work with the Method thesis is introduced during the second part of the course, but the most part of the work will be conducted during the last three weeks of course work. The Method thesis is examined by handing in the written report and by orally presenting the report at the final day of the course.

Learning outcomes

After completing the course, the students shall reach the following learning outcomes:

Knowledge and understanding

- Display knowledge and understanding of how to design social science research projects in order to make causal inferences.
- Display knowledge and understanding of the empirical application of statistical research methods for causal inference aimed to answer various research questions in the social sciences.
- Display comprehensive knowledge about the usage of different statistical techniques, primarily multivariate regression techniques.

Competence and skills

- Be able to independently use methods of data management and to apply various statistical techniques on different data sources using a statistical computer package.
- Be able to independently design and statistically test theoretically deduced causal models.
- Demonstrate good ability to correctly present, visualize, interpret and analyze empirical results obtained through different statistical methods.
- Independently plan and carry out a smaller study to solve a causal inference problem using appropriate statistical methods in a research paper.

- Independently produce text in accordance with good academic practice, including proper citation technique and use of references
- Communicate clearly and proficiently in English both orally and in writing

Judgement and approach

- Independently and critically assess research designs, methodology, empirical analyses and conclusions from social science research based on statistical analysis.
- Independently and critically assess and evaluate principles of operationalization, validity and reliability.
- Identify, evaluate and judge problems of research ethics in Social Sciences.

Examination and course requirements

The course is examined by completing one (1) oral group assignment (1 hec), five (5) written group assignments (1 to 3 hec each, in total 8 hec), and by independently writing and orally presenting a Method thesis (6 hec).

To be awarded Pass, assignments 1-6 must be awarded Pass and assignment 7 (the Method thesis) must be awarded Pass.

To be awarded Pass with Distinction, assignments 1-6 must be awarded Pass and assignment 7 (the Method thesis) must be awarded Pass with Distinction.

To reach *Pass*, the assignments must be correctly carried out, you must understand the application of the statistical method(s) in question, you must present your work according to research standards and you must draw correct conclusions.

To reach *Pass with Distinction*, apart from the requirements for *Pass*, you also need to adopt an independent approach in the Method thesis including a well-motivated theoretical model, critically assess the chosen course of action by actively discussing strengths and weaknesses regarding the causal inference design, operationalization, analytic techniques, conclusions and alternative procedures.

We grade and provide written feedback on all written assignments through Canvas. For assignment 2-6, the grading teacher could comment directly in your submitted assignment on Canvas, and/or summarize the feedback in a comment to your assignment. For assignment 7, we use an assessment sheet (see the end of this document) which is uploaded on Canvas as feedback for the student.

Course work overview

Assignment #	Assignment topic	Examination form	Credits	Instructor
1	Design for causal inference	Oral group assignment: presentation at compulsory seminar	1	Mattias Agerberg
2	Linear regression I: Introduction	Written group assignment	2,5	Florence So
3	Linear regression II: Heterogenous effects	Written group assignment	1,5	Maria Solevid
4	Applied methods for causal inference I: Panel regression	Written group assignment	1	Rasmus Broms
5	Applied methods for causal inference II: Difference-in-difference	Written group assignment	1,5	Rasmus Broms
6	Applied methods for causal inference III: Synthetic controls	Written group assignment	1,5	Mattias Agerberg
7	Method Thesis	Written and oral individual assignment: paper and presentation at compulsory seminar	6	All instructors (Maria Solevid is the coordinating instructor)

Assignment 1

Instructions for assignment 1 will be available on CANVAS at the start of the course. The assignment, which is conducted in groups, consist of making an oral presentation of a causal inference design. **You are responsible for forming the 2-3 persons groups yourselves.** Generally, you work in the same group throughout assignment 1 to 6. Those who don't find a group, contact Maria Solevid.

Assignment 2-6, tutorials and groups

Instructions for assignment 2-6 will be available on CANVAS at the start of the course. All assignments are handed in on CANVAS (see instructions below). The responsible instructor uploads all necessary documents, such as Power Point presentation, do-files, and sometimes also data sets.

Prior to the **Stata tutorials for assignment 2-6** you must watch and follow the recorded lecture with the demonstration of the Stata procedures necessary to run the statistical analyses for the assignment. We strongly encourage you to apply the procedures while following the recorded lecture.

At the actual tutorial, you and your group (2-3 students) will work with the assignment with the possibility to receive hands-on help and tips from the teacher.

Assignment 7, The Method Thesis

The last three weeks of the course work are devoted to independent work. The Method thesis involves planning the carrying out of a smaller study to solve a causal inference problem using appropriate statistical methods in a research paper. The method thesis resembles a master's thesis although much shorter and with empirical and methodological emphasis. To support

your work with the Method thesis, one of the instructor will be assigned as your advisor and you will be able to receive feedback on your initial idea (the abstract) and on your draft. We also offer a short Stata help desk session during the work with the method thesis. The method thesis will be presented by each student at the final seminar. Read more about the method thesis in the Assignment 7, method thesis instructions, uploaded on Canvas.

Late/revised/failed assignments

Assignment 1

Students not participating in the seminar and the oral presentation need to complete a supplementary assignment (a written assignment) within **two weeks**. Students who Fail the oral assignment or the supplementary assignment also need to hand in a written assignment, by the latest at the last day of the course (January 16, 2026). See assignment instructions. We offer the possibility of extra-ordinary submission on March 13 and June 7, 2026.

Assignment 2-6

If your assignment is marked as *revisions needed*, the workshop instructor will specify the necessary revisions in the written feedback on CANVAS. Submit your revisions within **one week** after the feedback has been uploaded on CANVAS. If you are unable to submit within one week, the next deadline to submit revised assignments 2-6 for them to be considered completed during the Fall semester 2024 is January 16, 2026.

If your assignment is marked as *Fail*, you will need to revise and resubmit the workshop assignment according to the feedback provided. Hand in your work by the latest on the last day of the course (January 16, 2026) to be considered completed during the Fall semester 2024.

Regardless of whether you previously did not hand in an assignment or in the event you have failed an assignment, we offer the possibility of extra-ordinary submission on January 16, March 13 and June 7, 2026.

Assignment 7, Method thesis

If your Method thesis is graded ***revisions needed***, to achieve Pass, you'll need to revise your Method thesis. The responsible grader will specify the necessary revisions in the assessment sheet. The revised Method thesis must be handed in within **two weeks** after feedback has been sent to you. If you fail to hand in the revised version of the Method thesis on time, or if the revisions are not assessed as sufficient, the Method thesis will be graded as Fail.

If your Method thesis is marked as ***Fail***, you need to rewrite the Method thesis and hand it in again. Dates for extra-ordinary submission are March 13 and June 7 (2026).

If you do not hand in the Method thesis on January 15, 2026 and/or if you hand in but are not able to be present at the compulsory seminar January 16, you must record your presentation and complete a supplementary assignment. See Assignment 7, method thesis instructions on Canvas.

Please note that the instructors do not offer advice/feedback of your work in progress if you hand in your Method thesis after January 15, 2026.

Literature

If you cannot access the articles through the hyperlinks, search them through ub.gu.se

Books

Angrist, Joshua & Jörn-Steffen, Pischke. 2014. *Mastering metrics. The path from cause to effect*. Princeton, NJ: Princeton university press. 289 pages

Book excerpts

King, Gary, Robert O. Keohane & Sidney Verba. 1994. *Designing Social Inquiry. Scientific Inference in Qualitative Research*. Princeton, NJ: Princeton University Press. Please note only chapter 1 is included as reading. A 15-page excerpt from chapter 1 "The Science in Social Science" is uploaded on Canvas.

Articles, required reading

Abadie, Alberto et al. 2015. "[Comparative Politics and the Synthetic Control Method](#)", *American Journal of Political Science*, 59(2): 495-510.

Adcock, Robert & David Collier 2001. "[Measurement Validity: A Shared Standard for Qualitative and Quantitative Research](#)". *American Political Science Review*, 95(3): 529-546.

Brambor, Thomas et al. 2006. "[Understanding Interaction Models: Improving Empirical Analysis](#)". *Political Analysis*, 14(1): 63-82.

Gerber, Alan et al. 2014. "[Why People Vote: Estimating the Social Returns to Voting](#)", *British Journal of Political Science*, 46: 241-264.

Hainmueller, Jens. & Dominik. Hangartner. 2019. "[Does Direct Democracy Hurt Immigrant Minorities? Evidence from Naturalization Decisions in Switzerland](#)", *American Journal of Political Science*, 63(3): 530-547.

Keele, Luke. 2015. "[The Statistics of Causal Inference: A View from Political Methodology](#)", *Political Analysis*, 23: 313–335.

Articles, optional reading

Bove, Vincenzo et al. 2023. "[Are the Effects of Terrorism Short-Lived?](#)", *British Journal of Political Science*, Forthcoming: 1-10.

Corbacho, Ana et al. 2016. "[Corruption as a Self-Fulfilling Prophecy: Evidence from a Survey Experiment in Costa Rica](#)", *American Journal of Political Science*, 60(4): 1077-1092.

Gerber, Alan et al. 2008. "[Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment](#)", *American Political Science Review*, 102(1): 33-48.

Holland, Paul. W. 1986. "[Statistics and Causal Inference](#)", *Journal of the American Statistical Association*, 81(396): 945-960.

Martinangeli, Andrea et al. 2023. "[Institutional Quality Causes Generalized Trust: Experimental Evidence on Trusting under the Shadow of Doubt](#)", *American Journal of Political Science*, Forthcoming: 1-16.

Web resources

Angrist, Joshua (2021) Prize lecture <https://www.nobelprize.org/prizes/economic-sciences/2021/angrist/lecture/>

Mastering metrics, website, <https://www.masteringmetrics.com>

UCLA, Stata resources <https://stats.oarc.ucla.edu/stata/>

Additional material (PowerPoint presentations, assignment instructions, video tutorials) will be available on CANVAS.

Schedule

The below schedule is *not* updated throughout the course = *always use the online schedule on Canvas* for exact dates and for information about time and room. Please also bring laptops to all tutorials.

Submission deadlines highlighted in yellow.

Date	Week 45	Reading material/comment
5/11	L1 Course introduction (MS)*	Read the course guide, browse the course page on Canvas and mark down the important dates and deadlines.
5/11	L2 Recap: Basic statistical analysis (FS)	Adcock & Collier, 2001, King, Keohane & Verba, 1994 (on Canvas)
6/11	L3 Design for causal inference I: Potential outcomes (MA)	<u>Required reading:</u> Angrist & Pische, Introduction & ch 1, Keele, 2015 <u>Optional reading:</u> Holland, 1986
7/11	L4 Design for causal inference II: Experiments (MA)	<u>Required reading:</u> Angrist & Pische, 2014 Introduction ch 1, Gerber et al., 2014, Hainmueller & Hangartner, 2019 <u>Optional reading:</u> Bove et al., 2023, Corbacho et al., 2016, Gerber et al., 2008, Martinangeli et al., 2023
Date	Week 46	
10/11	Hand in Assignment 1 (presentation slides) Design for causal inference*	Assignment instructions on Canvas
11/11	Oral presentation, Assignment 1 Design for causal inference (MA)*	Compulsory attendance and active participation at seminar
12/11	Stata tutorial: Introduction to Stata (FS)	Bring your laptop!
13/11	L5 Linear regression I: Introduction I (FS)	Angrist & Pische 2014, ch 2 Six short videos found on https://www.masteringmetrics.com/online-metrics-resources/ (A world of potential (outcomes, that is) and Refreshing regression)
13/11	L6 Linear regression I: Introduction II (FS)	Angrist & Pische 2014, ch 2 Videos from Mastering metrics (see above)
14/11	Stata tutorial, Assignment 2 part 1 (FS)	Bring your laptop! Watch the Stata demonstration video before the tutorial. Assignment instructions on Canvas
Date	Week 47	
17/11	L7 Linear regression I: Introduction III (FS)	Angrist & Pische 2014, ch 2 Videos from Mastering metrics (see above)
17/11	L8 Linear regression I: Introduction IV (FS)	Angrist & Pische 2014, ch 2 Videos from Mastering metrics (see above)
18/11	Stata tutorial, Assignment 2 part 2 (FS)	Bring your laptop! Watch the Stata demonstration video before the tutorial. Assignment instructions on Canvas
21/11	Hand in Assignment 2 Introduction to linear regression on Canvas by the latest 16:00*	
	Week 48	
24/11	L9 Linear regression II: Heterogenous effects I (MS)	Brambor et al, 2006 UCLA, Decomposing, probing, and plotting interactions in Stata
26/11	Stata tutorial, Assignment 3 (MS)	Bring your laptop! Watch the Stata demonstration video before the tutorial. Assignment instructions on Canvas

28/11		Hand in Assignment 3 Heterogenous effects in linear regression on Canvas by the latest 16:00*	
	Week 49		
1/12	L10	Applied methods for causal inference I: Panel regression (RB)	Hainmueller & Hangartner, 2019
2/12		Stata tutorial, Assignment 4, (RB)	Bring your laptop! Watch the Stata demonstration video before the tutorial. Assignment instructions on Canvas
4/12		Hand in Assignment 4 Panel regression on Canvas by the latest 16:00*	
5/12		Independent work with preparations for method thesis and upcoming abstract submission!	
	Week 50		
8/12	L11	Applied methods for causal inference II: Difference-in-difference (RB)	Angrist & Pische 2014, ch. 5
9/12		Stata tutorial, Assignment 5, (RB)	Bring your laptop! Watch the Stata demonstration video before the tutorial. Assignment instructions on Canvas
12/12		Hand in assignment 5 Difference-in-difference on Canvas by the latest 16:00*	
	Week 51		
15/12	L12	Applied methods for causal inference III: Synthetic control (MA)	Abadie et al., 2015
16/12		Hand in Abstract for the Method thesis on Canvas by the latest 09:00	It's required to hand in the abstract to be assigned an advisor and to receive feedback both on the abstract and the upcoming draft
17/12		Stata tutorial Assignment 6 (MA)	Bring your laptop! Watch the Stata demonstration video before the tutorial. Assignment instructions on Canvas
19/12		Hand in assignment 6 Synthetic control on Canvas by the latest 12:00	
19/12		Individual Q&A session related to the Method thesis (MA, RB, MS, FS), Zoom	ZOOM Appr. 15 minutes/student. Exact time and Zoom link will be sent out by your advisor.
	Week 2		
8/1		Hand in Method thesis draft on Canvas by the latest 09:00	
9/1		Individual supervision meeting on Zoom, discussion of Method thesis drafts (MA, RB, MS, FS)	ZOOM Appr. 30 minutes/student. Exact time and Zoom link will be sent out by your advisor.
	Week 3		
15/1		Hand in Assignment 7 Method thesis on Canvas by the latest 12:00*	
16/1		Seminar; Presentation of Method thesis (MA, RB, MS, FS)	Compulsory attendance (presentation and active participation)!

A note on academic writing

Throughout your time at University you will write a number of academic texts (such as analytical papers and literature reviews, home exams and the final thesis). What follows are some of the basic—and hugely important—principles for independent academic writing:

Academic writing is based on the principle of clarity and transparency regarding the origin of the information and knowledge conveyed by the text. Academic texts are characterized by independent and self-designed writing.

In combination, the above principles mean that in academic writing we use the ideas, thoughts, arguments, opinions, conclusions of others, but we make clear the origin of these by quoting and referring. We communicate with, build on, and credit previous research, authors, sources, **but** we do not copy them! Copying is plagiarism, which is the same as cheating (see more below).

Some examples of plagiarism and cheating: 1) Copying an existing text. 2) Moving and deleting a few words from an existing text, 3) Translating an existing text from another language, 4) Using an AI to generate your text.

There should be no doubt whatsoever in the text about which ideas, thoughts, arguments, opinions and conclusions are yours (absent a reference) and which arguments you obtained from other texts (reference to the author included). There are different reference styles (see ulr below) but a good rule of thumb, for maximum transparency, is to include the reference in the text, like this: (author surname, year, page), i.e. (Sikkink, 2000, 13).

Here are ulrs that further describe the essentials of academic writing (please click “In English” in case the page appears in Swedish):

Searching and citing

<http://www.ub.gu.se/skriva/>

Web course: Academic integrity, see the course page on Canvas! You can also read more at <https://pil.gu.se/resurser/plagiering>.

Resources for Academic Writing (ASK)

<https://studentportal.gu.se/english/help-and-guidance/academic-Language-and-writing>

Attempts to deceive during examinations

An attempt to deceive during examinations means that a student has attempted to give an impression of their performance in a written exam or some of form of assessment that is not true. That is what is commonly referred to as cheating or attempting to cheat. There are different kinds of attempts to deceive, such as plagiarism, prohibited aids and prohibited collaboration. Deception can also take other forms.

- **Plagiarism** means that a student has presented someone else’s text or other material as if it were their own text or material.
- **Prohibited aid** means that a student has used aids, including AI, that were not permitted according to the instructions.
- **Prohibited collaboration** means that students have collaborated with each other in breach of the instructions.

Plagiarism: Copying material from the Internet, from books, journals, reports, or from a fellow student and then presenting it as one's own is strictly forbidden! Plagiarism is theft of someone else's work and will result in immediate fail of the assignment. Please note that the inclusion of a source in a bibliography list is not in itself sufficient attribution of another's work. In addition, unless it is a proper citation with a reference, any direct copying of text is forbidden. All submitted student work is screened with a program called *Ouriginal*, which compares a student's written work with published materials (including everything published on the internet) and all the work from prior students, to detect plagiarism. The material you have read and based your arguments and factual statements on should always be referenced using proper referencing technique. More information on references and how to avoid plagiarism can be found here: <https://studentportal.gu.se/english/help-and-guidance/search-read-write/reference-management-and-plagiarism>

Prohibited aid: All your submitted texts should be your own original writing, and not written by someone else or generated by AI. Unless there are specific instructions for the use of AI for an assignment, it should not be used and presented as your own text. The use of any tools to cheat during on campus exams (e.g. DISA) is also strictly forbidden.

Prohibited collaboration: students can discuss their work, but individual submissions must be your own original writing, and hence not too similar the work of others. Be extra careful to avoid using common notes from such collaborations.

Any kind of cheating on examinations is considered a serious offence. Proved or suspected cheating will be reported to the Disciplinary Committee of the University of Gothenburg, in accordance with the procedure of disciplinary matters at the department of political science.

The consequences may be severe for a student if found guilty. A student may be suspended from classes and campus for a determined time-period (from 3 weeks to half a year), which will result in a temporary suspension of all financial aid (including CSN) and may jeopardize a student's residence permit status. More information on disciplinary matters, including plagiarism and its consequences, can be found here: <https://studentportal.gu.se/english/study-environment-and-rules/disciplinary-matters/>

AI policy for SF2321

At the end of all assignments, after the references but before the appendix, you should include a declaration of AI content that describes if and how AI tools were used.

On SF2321, it **is permitted** to use generative AI (such as ChatGPT) for the following purposes:

- To troubleshoot or generate Stata code
- To get help with interpretations of results (for example regression coefficients)
- Better understand advanced statistical procedures
- Get ideas for topics

It is **forbidden** to use generative AI tools for the following purposes:

- Generate text that you copy/paste straight into your document – this is presenting someone else's text as your own and by default plagiarism.

Examples of text that may not be copy/pasted from AI tools are typical paper/thesis elements such as introduction, research question, literature review, hypotheses, descriptions of variables, descriptions of method, results interpretations, conclusion.

- Translating purposes – use rule-based AI tools instead for language/grammar help
- Upload data and conduct actual data analysis. You use Stata for these purposes.

Always remember that generative AI, such as ChatGPT, can lie. It may suggest Stata code that doesn't work, make up references that doesn't exist and so on. AI generated texts tend to be unnecessary long and more unfocused, including many fancy words that doesn't really mean anything. Sentences also tend to vary less in length and structure.

An academic text should always be straightforward and to the point. The text that you submit must 100% be something you stand behind yourself and that you understand every part of. You can never blame wrong assumptions or facts, fuzzy sentences, wrong interpretations of coefficients, or flawed Stata code, on AI.

About equal treatment

The University of Gothenburg aims to promote equal rights and opportunities for students and employees and to combat discrimination related to gender, gender identity or gender expressions, ethnicity, religion or other beliefs, sexual orientation, disability or age. If, however, you as a student at the Department of Political Science become subject to such acts and actions, we encourage you to contact us to enable the opportunity for appropriate action. You can contact: the course coordinator, director of studies Birgitta Niklasson, equal treatment representative Ulrika Möller, or student counselor Eva Nilsson. Student counselors have confidentiality. You can also turn to these persons if you experience problems or discomfort in the study environment, even if not targeted at you. All contact information is available on our website.

Grading form, Method thesis
SF2321, Applied Statistical Analysis

Name of student:	
Title of Method thesis:	
Criteria	Assessment
a) There is an interesting and, from a social science perspective, relevant introduction presenting a well-motivated causal research question	
b) Relevant theory and previous research findings connected to the research question are presented in a short and distinct literature review	
c) The causal effect of interest is formalized and described using the potential outcomes framework	
d) The formulated hypotheses clearly state the expectations regarding the causal effect.	
e) The research design (the identification strategy and the data material) used to test the stated hypotheses is described, discussed, and assessed	
f) The relevant counterfactual and the main assumptions needed to identify the causal effect of interest are described.	
g) Central concepts are defined and operationalized, and the validity of the measurements is discussed	
h) The chosen statistical method is transparently described and motivated	
i) Principles of research ethics are discussed (if applicable)	
j) The results from the statistical analyses are presented and interpreted in a correct way	
k) The research question is answered, and the hypotheses are tested and assessed.	
l) The conclusion re-connects to the research question and includes suggestions of how further research on the topic could be conducted.	
m) Language, structure, table design and graph design are appropriate.	
m) The oral presentation distinctly illustrates the most important results from the Method thesis.	

Additional comments:
Grade:
<i>Examined by: Mattias Agerberg/Rasmus Broms/Maria Solevid/Florence So.</i>